

Agrawal, Gans & Goldfarb (2025)

The Economics of Bicycles for the Mind

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Repository 2 - Week of August 24–29, 2026

`github.com/amchavezu/ai-02-agrawal`

Question. How do cognitive tools change effort, productivity, and the value of implementation skill and judgment?

Single mechanism. A tool substitutes for implementation effort but can complement opportunity and payoff judgment.

Conditional on finding an opportunity, the agent chooses $e_t \geq 0$:

$$\max_{e_t \geq 0} M(e_t; \theta) = p(se_t; \theta)\alpha\Delta - c(e_t; \theta).$$

- ▶ s : implementation skill; α : payoff judgment.
- ▶ $\gamma(t)$: opportunity judgment; θ : tool quality.
- ▶ Interior FOC: $p'(se_t^*; \theta)s\alpha\Delta = c'(e_t^*; \theta)$.

Cognitive-tool conditions. For $\theta' > \theta$ and every e ,

$$p(se; \theta') \geq p(se; \theta), \quad c(e; \theta') \leq c(e; \theta),$$

while $p'(se; \theta)/c'(e; \theta)$ strictly decreases in θ for $e > 0$.

With increasing/concave p , increasing/convex c , and an interior optimum:

$$e_t^*(1) < e_t^*(0), \quad e_t^*(\theta) = e^*(\theta), \quad V_0(\theta) > V_0(0).$$

Intuition. Tool adoption raises achievable net benefit but lowers the marginal return to direct human implementation effort.

- ▶ Rewrote continuation value as $V(\theta) = H + \theta Z$, where $H = (\Delta^2/4)\Gamma\alpha^2s$ and $Z = \Gamma/s$.
- ▶ Derived $\text{Var}(V) = \text{Var}(H) + 2\theta \text{Cov}(H, Z) + \theta^2 \text{Var}(Z)$.
- ▶ Relaxed $\Gamma \perp s$: initial equalisation now requires $\mathbb{E}[\Gamma^2] < \mathbb{E}[\Gamma s]\mathbb{E}[\Gamma/s]$.
- ▶ Wrote a standard-library Python check with three distributions and reproducible CSV, text, and SVG outputs.

Counterexample output: condition (30) holds, but $\theta^* = 1.792$ and $d \text{Var}(V)/d\theta|_{\theta=1} = -0.0727$.

Loose claim. “Variance is U-shaped in AI quality.”

Audit. The object is $\text{Var}(V(\theta))$. Condition (30) makes the initial slope negative and implies $\theta^* > 0$, but a positive slope at $\theta = 1$ also requires

$$\theta^* < 1.$$

The paper’s Appendix A.3 proves a positive slope of the derivative, not that its zero occurs before one.

Verdict. The AI’s unconditional statement is incorrect, and the paper’s displayed endpoint claim is under-conditioned.

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Guide already included
in hand/.